Agentic AI Log Analysis System

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Quick Start

Generate Demo Logs (5,000)

Load Built-in Sample File

DB Records: 20 sessions | 20 incidents | 10216.0 KB

Multi-Agent Orchestration & ML Trans

Inspect inputs, outputs, execution latencies, and findings for each of the 6 specialized AI agents:

Log Parser Agent (1448.6 ms) — SUCCESS

1. Ingests raw .log, .txt, .csv streams, parses timestamps, levels, services, and endpoints.

Input Summary: Processed file 'synthetic_demo_logs.log' (5007 total lines, format: Unstructured/Standard

Output Summary: Successfully parsed 5007 events (0 malformed/skipped lines). Severity breakdown: CRIT 441

Key Findings & Conclusions:

- Parsed 5007 valid operational log entries from synthetic_demo_logs.log.
- Identified 7 unique applications/services.
- Severity distribution: 194 Critical, 333 Error, 441 Warn, 4039 Info.
- Detected 12 unique API endpoints.

Incident Detection Agent (9877.3 ms) — SUCCESS

2. Heuristic rules + Scikit-Learn TF-IDF classifier to detect anomalies and assign severities.

Input Summary: Scanned 5007 log events across rules and ML models.

Output Summary: Detected 831 anomalous/incident log events. Breakdown by severity: 293 Critical, 501 H

Key Findings & Conclusions:

- Flagged 831 incident candidate events using 7 heuristic rules and TF-IDF ML model.
- Top incident categories: HTTP_SERVER_ERROR (326), SUSPICIOUS_AUTH (141), DATABASE_FAILURE (1
- ML Classifier Performance: Accuracy=70.00%, F1=72.73%, Precision=66.67%.
- Critical-tier events requiring immediate correlation: 293.

Supervised ML Classifier Validation Metrics

Model Accuracy	Precision	Recall
70.00%	66.67%	80.00%

Architecture: TF-IDF + LogisticRegression (Scikit-Learn) | Evaluated on holdout test set with stratified sampling.

Event Correlation Agent (453.6 ms) — SUCCESS

3. Groups related anomalous events using sliding temporal windows, endpoints, and trace links.

Input Summary: Correlated 831 candidate detections against 5007 total events.

Output Summary: Consolidated into 6 distinct operational incidents (6 Critical, 0 High). Found 4 cross-serv

Key Findings & Conclusions:

- Clustered 831 anomaly signals into 6 actionable incidents.
- Average correlated events per incident: 138.5 events.
- Detected 4 cross-service failure cascades (e.g. downstream API failing due to upstream DB).

- Temporal window used for correlation: ±5 minutes.

Root Cause Investigation Agent (9.7 ms) — SUCCESS

4. Reconstructs causal event sequences to formulate evidence-based root cause hypotheses.

Input Summary: Investigated causal telemetry for 6 active incidents.

Output Summary: Formulated 9 evidence-backed root cause hypotheses across 6 incidents (3 High confide

Key Findings & Conclusions:

- Synthesized 9 possible root cause scenarios supported by extracted log evidence lines.
- High confidence hypotheses: 3 (corroborated by clear stack traces or DB connection logs).
- Uncertain/Sparse evidence cases: 6 (telemetry lacks upstream diagnostic context).
- Ethical Observability Standard: All conclusions remain strictly designated as 'Possible Causes' awaiting

Resolution Recommendation Agent (0.1 ms) — SUCCESS

5. Recommends tactical containment steps, diagnostic commands, and preventive runbooks.

Input Summary: Generated troubleshooting runbooks for 6 active incidents.

Output Summary: Prescribed 24 actionable mitigation steps across 6 incidents (12 immediate containment commands).

Key Findings & Conclusions:

- Formulated 24 prioritized troubleshooting recommendations.
- Immediate containment actions: 12 (focused on preventing service cascading failure).
- Included copy-ready shell, SQL, and CLI diagnostic commands for on-call engineers.
- Policy: Actions are recommended guidance; platform requires engineer verification before marking res

Incident Report Agent (1.7 ms) — SUCCESS

6. Compiles comprehensive post-mortem audit reports in TXT, CSV, and JSON formats.

Input Summary: Processed session 'SESS-273FBEE5' containing 6 incidents.

Output Summary: Compiled audit-grade reports across 3 formats (TXT: 17000 chars, CSV: 7 rows, JSON: 44

Key Findings & Conclusions:

- Generated executive text summary with operational metrics and troubleshooting runbooks.
- Produced structured CSV incident registry for ITIL/SIEM record ingestion.
- Generated full JSON artifact containing complete agent findings, evidence, and ML evaluation stats.